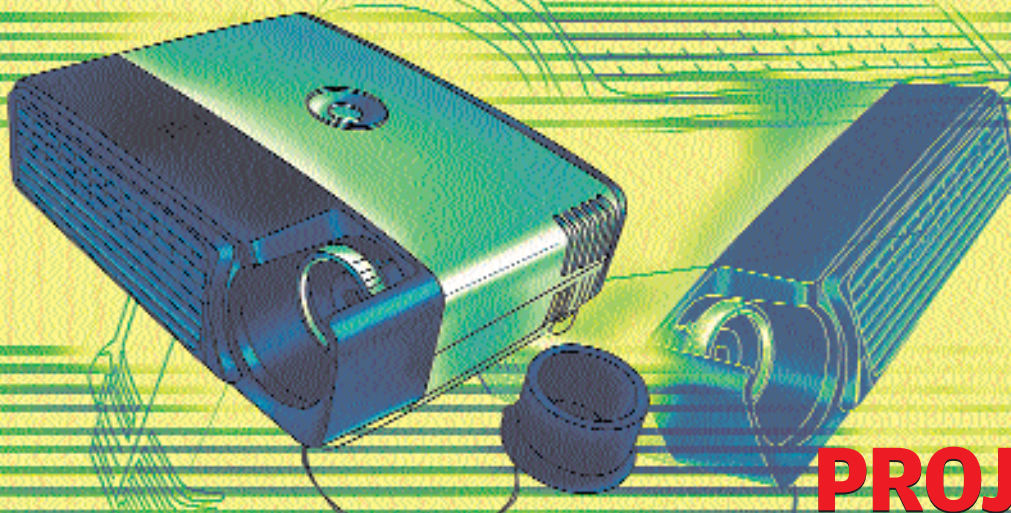




PROJECTORS

Video projectors come in two flavours—tubed and LCP (Liquid Crystal Projection). Tubed projectors are widely used in cinemas and public displays. The LCP type of projectors are again classified as standard LCD, polysilicon LCD and DLP (digital light processing)—the latter is fast gaining prominence. The image quality of tubed projectors is better than that of LCP, but tubed projectors are much heavier.

Questions to ask the dealer

Does it have a zoom lens?

A zoom lens allows you to adjust the size of the projected image, which is very convenient and allows you to project on screens of various sizes and at varying distances.

What are the video formats supported?

The projector should support a minimum of three video formats—NTSC, PAL and SECAM.

Does it have ceiling mount support?

Verify if the projector can be mounted on the ceiling, and also if it is capable of projecting images upside down.

Can it be used with a Mac?

PC and PC compatibles should not have much of a problem in this regard, but if you are using a Macintosh, it requires a

separate connector or adapter. Ask about this connector.

What extra features are available?

Look for features such as power zoom, a remote control, and input ports such as DVI, BNC, and S-Video and audio.

Usage tips

❑ Avoid using an LCD projector at a stretch for more than 10 hours. The heat will cause the quality of the LCD to deteriorate, and may even cause permanent damage to the LCD.

❑ In order to achieve optimal viewing quality, increase the brightness of the projector in places where total darkness cannot be achieved.

❑ Always remember that the projector will do its job best when its resolution matches that of the computer you're connecting it to. So, if you're using a projector with SVGA resolution, set your computer to 800 x 600 resolution.

❑ Never rely on the Internet connection for presentation of a Web site! Download the pages beforehand, or take screenshots of the Web pages and make slides from these.

Interesting Facts

DLP technology is based on an optical

semiconductor chip known as the DMD (Digital Micromirror Device), which is made up of millions of tiny mirrors that can rotate at an angle of 10 to 12 degrees. These mirrors can be switched on and off thousands of times per second, and direct light towards and away from the pixel space.

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DLP projectors are generally believed to provide smoother images and better colour accuracy, and have lower power consumption as well as higher contrast and brightness than LCD projectors.

CRT projectors aren't very often used, and we haven't even mentioned them thus far. As it happens, the image quality offered by a CRT projector is tough to beat! The problem here is that the cost is very high, the projector is too large, and the calibration needs expertise.

Points To Ponder

How bright should the projector be?

The brightness of a projector is specified in ANSI lumens. Typically, small conference rooms or classrooms require less than 1000 lumens. Large conference rooms, say, may require 2000 to 3000 lumens, whereas auditoriums, theatres and the like require projectors with a brightness above 3000 lumens.

What should the resolution be?

The various resolutions available are SVGA (800 x 600), XGA (1024 x 768), SXGA (1280 x 1024) and UXGA (1600 x 1200). SVGA is perfectly OK if presentations are what you are

going to use the projector for. XGA is suitable for projecting finer images such as spreadsheets, where numbers and letters need to be legible. For very fine images such as engineering drawings, and high-resolution maps, you might need SXGA or even UXGA, depending on the resolution of the image to be projected.

What about the weight and portability?

If you travel, and make presentations on the go, the projector should be light enough to carry around. If you only intend to make presentations at your office, for example, a heavier projector should be just fine.

The Future

Various new types of projector technologies are being developed or are nearly available. LCOS (Liquid Crystal On Silicon wafer) is a combination of the LCD and DLP technologies. It has high pixel density, which results in high-resolution projectors that are especially good for multimedia applications. LCOS resolution starts at SXGA. Then there is holographic projection, in which a projector shines on a thin holographic screen attached to a semi-transparent surface such as plexi-glass—and the image appears to float in mid-air! ■